

BEND, OR, USAWMO: **720638**Lat: **44.095N**Lon: **121.200W**Elev: **1055**StdP: **89.27**Time Zone: **-8.00 (NAP)**Period: **09-19**WBAN: **00224****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -14.7	(c) -11.3	(d) -18.7	(e) 0.8	(f) -5.7	(g) -17.0	(h) 1.0	(i) -4.0	(j) 12.2	(k) 7.8	(l) 10.9	(m) 7.2	(n) 1.5	(o) 340	(p) 0.576

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 18.2	(c) 33.6	(d) 15.3	(e) 32.2	(f) 15.1	(g) 30.0	(h) 14.7	(i) 16.8	(j) 29.1	(k) 16.1	(l) 28.7	(m) 15.3	(n) 27.6	(o) 3.9	(p) 0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.6	10.4	19.6	12.0	9.9	18.8	10.2	8.8	18.2	51.0	28.9	48.7	29.3	46.4	27.9	20.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 9.5	(b) 8.2	(c) 7.3	(e) -19.7	(f) 36.3	(g) 4.7	(h) 1.6	(i) -23.1	(j) 37.5	(k) -25.9	(l) 38.4	(m) -28.5	(n) 39.4	(o) -31.9	(p) 40.5		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.3	1.5	2.1	4.8	7.1	11.7	15.0	20.1	19.9	15.4	9.2	3.9	0.0
	DBStd	8.10	4.85	4.66	3.80	4.01	4.29	4.06	3.61	3.54	4.27	3.95	5.19	5.15
	HDD10.0	1366	266	223	165	106	31	6	0	0	6	63	190	311
	HDD18.3	3497	523	456	420	336	208	114	24	24	106	285	434	568
	CDD10.0	1102	1	1	3	20	86	157	314	308	169	37	6	1
	CDD18.3	192	0	0	0	0	3	15	81	74	18	1	0	0
	CDH23.3	3464	0	0	0	18	114	341	1387	1172	407	26	0	0
	CDH26.7	1389	0	0	0	2	21	100	626	499	138	4	0	0
(14) Wind	WSAvg	2.7	2.5	2.9	3.1	3.0	2.6	2.6	2.5	2.5	2.4	2.6	2.8	2.6
(15) Precipitation	PrecAvg	290	45	26	22	17	22	19	13	14	10	16	39	50
	PrecMax	487	111	93	70	55	104	81	95	70	44	55	116	222
	PrecMin	138	3	2	0	0	0	0	0	0	0	0	2	0
	PrecStd	75	26	20	16	13	22	19	17	16	10	11	33	43
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.0	16.3	19.9	26.0	28.9	32.4	36.0	36.3	32.7	27.2	19.8	12.8
		MCWB	6.9	8.3	8.5	11.9	13.8	16.7	16.1	15.8	14.4	12.0	9.6	6.8
	2%	DB	12.3	13.0	17.4	22.2	26.4	29.1	33.8	33.0	30.0	22.6	16.3	11.1
		MCWB	6.3	6.5	7.3	10.4	12.8	15.2	15.5	14.8	14.0	10.4	9.0	6.4
	5%	DB	9.9	11.1	14.0	18.7	23.9	27.4	32.4	32.0	27.8	19.8	13.7	8.8
		MCWB	5.2	5.6	6.5	9.1	11.7	13.8	15.2	14.7	13.4	9.8	7.7	5.4
	10%	DB	7.8	8.9	12.2	15.2	22.0	25.0	30.2	29.1	25.2	17.3	11.4	7.3
		MCWB	3.9	4.8	6.0	7.0	11.0	12.6	14.8	14.4	12.5	9.2	6.6	4.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.2	9.1	10.7	12.7	15.1	18.1	18.2	17.4	16.7	13.4	11.3	8.6
		MCDB	13.8	15.0	16.7	22.5	24.9	28.9	30.7	30.1	28.6	22.4	16.9	12.0
	2%	WB	6.8	7.7	8.8	10.8	13.8	16.3	16.8	16.5	15.4	11.8	10.0	6.8
		MCDB	10.6	13.0	14.1	20.5	23.6	27.5	29.6	29.3	26.1	19.6	15.0	9.7
	5%	WB	5.4	6.4	7.3	9.4	12.7	14.8	16.1	15.8	14.3	10.7	8.6	5.4
		MCDB	9.2	9.9	12.7	17.2	21.2	25.0	29.4	28.2	25.7	17.7	13.8	8.4
	10%	WB	4.2	4.9	6.2	8.0	11.6	13.5	15.3	15.1	13.3	9.7	6.7	4.0
		MCDB	7.7	8.6	11.5	14.7	19.8	22.9	28.7	27.1	23.9	16.3	10.7	6.8
(35) Mean Daily Temperature Range	MDBR		9.4	10.1	11.8	13.3	14.7	15.5	18.2	17.7	16.5	14.0	11.1	9.2
	5% DB	MCDBR	12.9	13.2	15.6	19.0	18.6	18.6	19.6	19.5	20.0	19.3	14.1	10.6
		MCWBR	7.9	7.9	8.2	8.7	7.4	7.2	6.6	6.5	8.1	9.3	7.9	6.7
	5% WB	MCDBR	11.3	11.5	14.0	17.1	16.0	17.1	18.4	18.0	18.1	15.3	12.8	9.3
		MCWBR	7.5	7.2	8.1	8.4	6.7	7.2	6.5	6.3	7.5	8.0	7.5	6.4
(40) Clear-Sky Solar Irradiance	taub		0.278	0.283	0.288	0.307	0.324	0.314	0.326	0.315	0.299	0.297	0.276	0.276
	taud		2.543	2.561	2.571	2.512	2.487	2.543	2.518	2.565	2.585	2.579	2.597	2.544
	Ebn at Noon		867	926	962	960	946	953	937	940	936	895	866	833
	Edh at Noon		66	77	86	100	106	100	102	94	84	74	62	60
(44) All-Sky Solar Radiation	RadAvg		1.59	2.49	3.69	5.16	6.30	7.31	7.80	6.71	5.14	3.21	1.83	1.29
	RadStd		0.19	0.30	0.31	0.44	0.47	0.40	0.32	0.35	0.27	0.26	0.18	0.19

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+90	+35

Nomenclature: See separate page